Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
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Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: *To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration*	**Mission:** *To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.*

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name and Signature of Student and Date

Sheikh.
21/02/26

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Session	2025-26 (EVEN)	Course Name	Competitive Coding Lab
Semester	6	Course Code	23IOT1605
Roll No	63	Name of Student	Sheikh Wasimuddin
Practical Number	DBMS Practice 1		

Dataset01: ProductSalesDatabase Product

Table

product_id	product_name	category	price
1	Laptop	Electronics	60000
2	Mobile	Electronics	30000
3	Chair	Furniture	5000
4	Table	Furniture	8000
5	Headphones	Electronics	4000

SalesTable

sale_id	product_id	quantity	sale_date
1	1	2	2023-01-10
2	2	5	2023-02-12
3	3	3	2023-02-15
4	1	1	2023-03-01
5	5	4	2023-03-10

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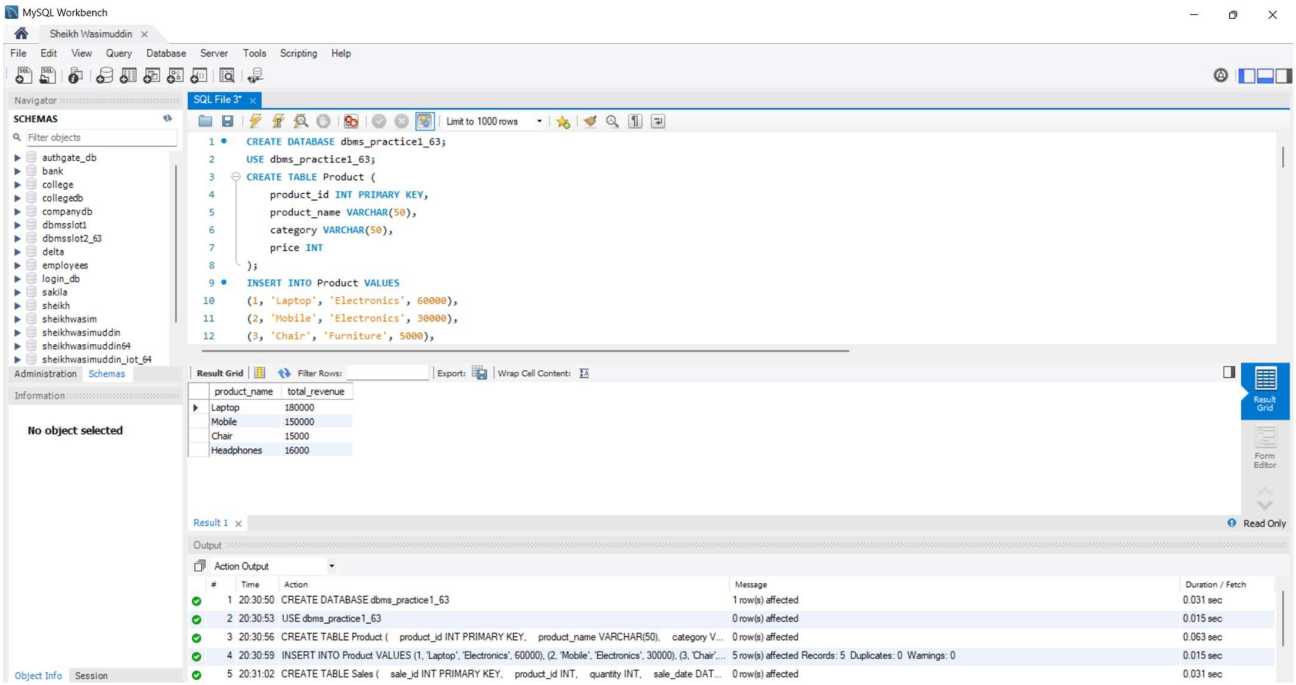

Practice Questions:

1. Find total revenue generated by each product.

```
SELECT p.product_name,
SUM(p.price * s.quantity) AS total_revenue
FROM Product p
JOIN Sales s ON p.product_id = s.product_id
GROUP BY p.product_name;
```

Result Grid | Filter Rows: | Export:

product_name	total_revenue
Laptop	180000
Mobile	150000
Chair	15000
Headphones	16000

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2. Display products that have never been sold.

```
SELECT p.product_name
FROM Product p
LEFT JOIN Sales s ON p.product_id = s.product_id
WHERE s.product_id IS NULL;


SELECT p.category,
SUM(p.price * s.quantity) AS total_sales
FROM Product p
JOIN Sales s ON p.product_id = s.product_id
GROUP BY p.category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

product_name
Table

3. Find category-wise total sales amount.

```
SELECT p.category,
SUM(p.price * s.quantity) AS total_sales
FROM Product p
JOIN Sales s ON p.product_id = s.product_id
GROUP BY p.category;

SELECT product_name, price
FROM Product
WHERE price > (SELECT AVG(price) FROM Product);
```

Result Grid | Filter Rows: | Export: | Wrap Cel

category	total_sales
Electronics	346000
Furniture	15000

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4. List products whose price is above average price.

```
SELECT product_name, price
FROM Product
WHERE price > (SELECT AVG(price) FROM Product);


SELECT p.product_name,
SUM(s.quantity) AS total_quantity
```

Result Grid | Filter Rows: | Export: | Wrap Cell

product_name	price
Laptop	60000
Mobile	30000

5. Find the most sold product (by quantity).

```
SELECT p.product_name,
SUM(s.quantity) AS total_quantity
FROM Product p
JOIN Sales s ON p.product_id = s.product_id
GROUP BY p.product_name
ORDER BY total_quantity DESC
LIMIT 1;


SELECT MONTH(sale_date) AS month,
```

Result Grid | Filter Rows: | Export: | Wr

product_name	total_quantity
Mobile	5

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6. Display month-wise totals sales.

```
SELECT MONTH(sale_date) AS month,
SUM(quantity) AS total_quantity
FROM Sales
GROUP BY MONTH(sale_date);



SELECT p.product_name,
COUNT(s.sale_id) AS times_sold
```

Result Grid | Filter Rows: | Export:

month	total_quantity
1	2
2	8
3	5

7. Find products sold more than once.

```
SELECT p.product_name,
COUNT(s.sale_id) AS times_sold
FROM Product p
JOIN Sales s ON p.product_id = s.product_id
GROUP BY p.product_name
HAVING COUNT(s.sale_id) > 1;


SELECT p.category,
SUM(p.price * s.quantity) AS total_revenue
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

product_name	times_sold
Laptop	2

8. Show highest revenue generating category.

```
SELECT p.category,
SUM(p.price * s.quantity) AS total_revenue
FROM Product p
JOIN Sales s ON p.product_id = s.product_id
GROUP BY p.category
ORDER BY total_revenue DESC
LIMIT 1;


SELECT *
```

Result Grid | Filter Rows: | Export: | Wra

category	total_revenue
Electronics	346000

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9. Find sales made in February only.

```
SELECT *
FROM Sales
WHERE MONTH(sale_date) = 2;
```

Result Grid | Filter Rows: | Edit:

sale_id	product_id	quantity	sale_date
2	2	5	2023-02-12
3	3	3	2023-02-15
NULL	NULL	NULL	NULL

10. Increase price of Electronics by 10%.

```
UPDATE Product
SET price = price * 1.10
WHERE category = 'Electronics'
LIMIT 100;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	20:38:17	UPDATE Product SET price = price * 1.10 WHERE category = 'Electronics' LIMIT 100	3 row(s) affected Rows matched: 3 Changed: 3 Warnings: 0	0.015 sec